

Introductory Statistics PS1

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] For observations [11, 11, 3, 6, 2], compute the sample mean and sample variance. Show deviations or an equivalent calculation.

2. [7 pts] A batch has 3 defective and 13 good units. Two are sampled without replacement. Compute $P(\text{both defective})$ and explain why the second denominator changes.

3. [8 pts] A sample of $n=40$ has $p=0.35$. Construct a 95% normal-approximation interval and state what the interval means.

4. [6 pts] A confidence interval is wide even though the point estimate is stable. Give two concrete changes that would narrow it and one trade-off.

5. [6 pts] A p-value is 0.03 for a pre-registered test at $\alpha=0.05$. State the decision and the conclusion without claiming that the null is proven false.

6. [8 pts] For observations [3, 6, 7, 3, 2], compute the sample mean and sample variance. Show deviations or an equivalent calculation.
